CO-3

# Probability

* Probability is the numerical measure of uncertainty.

## Applications

→ Weather forecasting.
→ Medical diagnosis.
→ Machine Learning.
→ Gaming.

$$P(E) = \frac{\text{No of favourable outcomes}}{\text{Total No of Possible outcomes}}$$

## Events

* A subset of the sample space.

## Simple Event

* Contains only one outcome.

## Compound Event

* Contains multiple outcomes.

## Complementary Events.

$$P(A') = 1 - P(A)$$

## Mutually Exclusive Events.

* Cannot occur simultaneously.

## Independent Events

* Occurrence of one event does not affect another.

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Dependent

* Occurrence of one event affects another.

Addition Rule $\Rightarrow P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Multiplication Rule $\Rightarrow P(A \cap B) = P(A)P(B)$.

Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0.$$

Bayes Theorem

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

Random Variables

* A random variable is a numerical quantity whose value depends on the outcome of a random experiment

$X: S \to R$

S = Sample space

R = Set of Real Numbers.

Discrete Random Variables

* Finite or countable values.

Continuous Random Variables

* Infinite values within an interval

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PMF (Probability Mass Function)

* PMF gives probabilities of discrete values.

$$P(X=x) \geq 0$$

$$\sum P(X=x) = 1$$

Variance (Discrete variables)

$$Var(X) = E(X^2) - [E(X)]^2$$

$$\sigma = \sqrt{Var(X)}$$

Probability Density Function (PDF)

$$f(x) \geq 0$$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$P(a < X < b) = \int_a^b f(x)\,dx$$

* For continuous variables.

Probability Distributions.

* How probabilities are assigned to outcomes of a random variable.

## Discrete Distributions

→ Bernoulli
→ Binomial
→ Poisson
→ Geometric
→ Negative Binomial

## Continous Distributions

→ Uniform
→ Normal
→ Exponential
→ Gamma
→ Beta.

### Bernoulli

* Used for experiments with two outcomes.

$$P(X=x) = p^x (1-p)^x$$

$$x \in (0,1)$$

Mean = p
Variance = p(1-p).

### Binomial

* Number of successes in n independent trials.

$$P(X=x) = \binom{n}{x} p^x (1-p)^{n-x}$$

mean = np
variance = np(1-p)

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## Poisson Distribution

* Used for counting events occurring over time.

Ex:- * Calls received per minute.
* Website visits.

$\lambda$ = Average rate.

$$P(X=x) = \frac{e^{\lambda} \lambda^{x}}{x!}$$

## Uniform Distribution

* All values have equal likelihood.

$$f(x) = \frac{1}{b-a}$$

→ Mean = $\frac{a+b}{2}$ ; Variance = $\frac{(b-a)^2}{12}$

a = max
b = min.

## Normal Distribution (Gaussian Distribution)

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$\mu$ = Mean
$\sigma$ = Standard Deviation.

## Exponential Distribution

* Models waiting times.

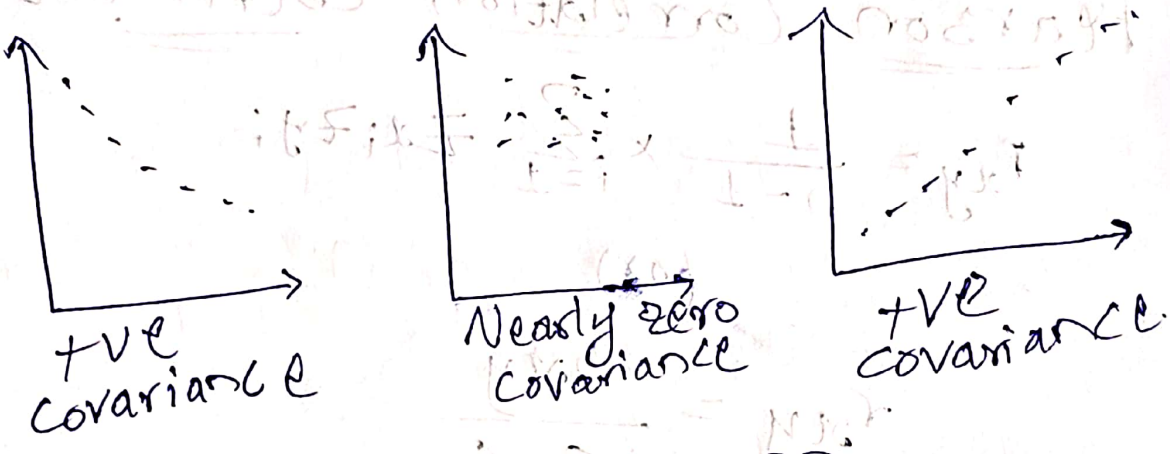
$\lambda$ = Rate. | $f(x) = \lambda e^{-\lambda x}$

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# Covariance

* Measure of relationship between 2 variables that is scale dependent.
i.e; how much will a variable change when another variable changes.

$$\text{Covariance}(x, y) = \sum \frac{(x_i - \bar{x})(y_i - \bar{y})}{N-1}$$




Ex:-
```
import numpy as np
x = np.array([70, 80, 90, 60, 75])
y = np.array([65, 85, 95, 55, 70])
cov_matrix = np.cov(x, y)
print(cov_matrix)
```

o/p:-
```
[[125.  175.]
 [175.  265.]]
```

## Correlation

* This overcomes the lack of scale dependency that is present in covariance by standardizing the values.

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{(N-1)(S_x)(S_y)}$$

## Pearson Correlation Coefficient

$$r_{xy} = \frac{1}{n-1} \times \sum_{i=1}^{n} z_{x_i} z_{y_i}$$

(or)

$$r_{xy} = \frac{S_{xy}}{S_x S_y}$$

0.8 - 1.0 → Strong Positive
0.5 - 0.8 → Moderate Positive
0.0 - 0.5 → Weak Positive
0 ⇒ No correlation
-0.5 - 0 → Weak Negative
-0.8 - -0.5 → Moderate Negative
-1.0 - -0.8 → Strong negative.

## Methods of studying correlation

1) Scatter Diagram Method

* To examine theories about cause-effect relationships & to search for root causes of an identified problem

2) Karl pearson's correlation coefficient

$$r = \frac{\text{Covariance}(x,y)}{S.D(x) * S.D(y)}$$

$$\text{Covariance}(x,y) = \frac{\Sigma xy}{N} - \left(\frac{\Sigma x}{N}\right)\left(\frac{\Sigma y}{N}\right)$$

$$S.D \text{ of } x = \sqrt{\frac{\Sigma x^2}{N} - \left(\frac{\Sigma x}{N}\right)^2}$$

$$S.D \text{ of } y = \sqrt{\frac{\Sigma y^2}{N} - \left(\frac{\Sigma y}{N}\right)^2}$$

(or)

$$r = \frac{\sum_{i=1} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\Sigma (x_i - \bar{x})^2 \Sigma (y_i - \bar{y})^2}}$$

* $-1 \leq r \leq 1$

t-Test ($n < 30$)

i) 95% confidence interval $\Rightarrow \bar{x} \pm t_{0.05} S/\sqrt{n}$

ii) 99% confidence interval $\Rightarrow \bar{x} \pm t_{0.01} S/\sqrt{n}$

$n > 30$

iii) 95% $\Rightarrow \bar{x} \pm 1.96 (\sigma/\sqrt{n})$

iv) 99% $\Rightarrow \bar{x} \pm 2.56 (\sigma/\sqrt{n})$

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## z-Table

| Level of Significance | Two tailed test | One-tailed test | |
|---|---|---|---|
| | | Right tailed | Left tailed |
| 0.10 (90% confidence) | ±1.645 | +1.28 | -1.28 |
| 0.05 (95%) | ±1.96 | +1.645 | -1.645 |
| 0.01 (99%) | ±2.58 | 2.33 | -2.33 |

## z-test

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

**Case 1:-** If $\sigma_1^2 = \sigma_2^2 = \sigma^2$ then

$$\mu_0 = \mu_1 = \mu_2$$

**Case 2:-**

$$\frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2}$$

**Case 3:-** If $\sigma_1^2 \neq \sigma_2^2$

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

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1) Null Hypothesis : $H_0$

2) Alternative Hypothesis:- $H_1$, this will enables us to decide whether we have to use a single-tailed test or two-tailed test.

3) Level of significance ($\alpha$).

4) Test $\Rightarrow$ $z = \frac{t - E(t)}{S.E(t)}$

Alternative Hypothesis

$H_1 : \mu \neq \mu_0$ (two-tailed)

$H_1 : \mu > \mu_0$ (right-tailed).

$H_1 : \mu < \mu_0$ (left-tailed).

Standard error = $\frac{\sigma}{\sqrt{n-1}}$

Decision Errors

1) Type I Error

* This occurs when the null hypothesis is rejected.

2) Type II Error

* This occurs when the null hypothesis is not rejected.

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## Chi-square Test

* Measures the alignment between two sets of frequency measures.

$$X^2 = \sum \frac{(O_i - e_i)^2}{e_i}$$

## ANOVA

*Analysis of Variance is a statistical tool for tests of significance.

## Regression Analysis

* A reasonable form of a relation between the dependent variable & the regressors x is the linear relationship:

$$Y = \alpha + \beta x$$

*Polynomial / Non-linear Regression.

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \cdots + \beta_n x^n$$

* Simple Linear Regression Model:

$$Y = \alpha + \beta x + c$$

→ $MAE = \frac{1}{N} \sum_{i=1}^{n} |y_i - \hat{y}_i|$

→ $MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y})^2$

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